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Automated lockout system for header

Abstract

Systems and methods for automatically configuring a cutterbar between a flexible configuration and a rigid configuration in response to actuation of a gauge wheel are disclosed. The cutterbar is coupled to the gauge wheel such that extension of the gauge wheel causes the cutterbar to move into the rigid configuration and retraction of the gauge wheel causes the cutterbar to move into the flexible configuration.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to an agricultural header and, particularly, to selectively configuring a cutterbar of an agricultural header.

BACKGROUND OF THE DISCLOSURE

(2) Agricultural harvesters use a variety of implements to gather crops. A “draper” or “draper header” is one such type of these implements. Conventional draper headers use conveyors with endless belts to carry cut crop material from leading-edge knives to center regions of the headers. From there, the cut crop material is conveyed into the harvesters. Once in the harvester, the cut crop material is further processed by separating grain from unwanted crop material (typically called “material other than grain” or “MOG”).

SUMMARY OF THE DISCLOSURE

(3) A first aspect of the present disclosure is directed to a system for automatically configuring a cutterbar of an agricultural implement. The system may include a cutterbar, a float arm engaged with the cutterbar, and an actuation system that connects the float arm and the gauge wheel. The float arm may be movable, via the actuation system, in response to one of retraction and extension of the gauge wheel to move the cutterbar between a rigid configuration and a flexible configuration. The actuation system may include a rocker arm that may include a pin and pivotable about a first axis. The rocker arm may be coupled to the float arm. The actuation system may also include a retainer and a cam coupled to the gauge wheel assembly. The retainer may include a recess that releasably receives the pin to selectively lock the pin into the recess and a protrusion. The cam may be moveable with the gauge wheel assembly to selectively engage the protrusion of the retainer to move the retainer between a locked configuration and an unlocked configuration.

(4) A second aspect of the present disclosure is directed to a method for moving a cutterbar of an agricultural implement between a rigid configuration and a flexible configuration in response to

articulation of a gauge wheel of the agricultural implement between a retracted configuration and an extended configuration. The method may include one of extending and retracting a gauge wheel of an agricultural implement and simultaneously one of moving the cutterbar into a flexible configuration in response to retraction of the gauge wheel and moving the cutterbar into a rigid configuration by an actuation system that is operated in response to retraction of the gauge wheel.

(5) A third aspect of the present disclosure is directed to an agricultural implement that automatically configures a cutterbar between a rigid configuration and a flexible configuration. The agricultural implement may include a frame; a gauge wheel assembly coupled to the frame and moveable between an extended position and a retracted position; a float arm pivotably coupled to the frame; a cutterbar engaged with the float arm, the cutterbar moveable between a rigid configuration and a flexible configuration in response to rotation of the float arm; and an actuation system that connects the float arm and the gauge wheel assembly. The float arm may be movable, via the actuation system, in response to one of retraction and extension of the gauge wheel to move the cutterbar between the rigid configuration and the flexible configuration. The actuation system may include a rocker arm comprising a pin and pivotable about a first axis, a retainer, and a cam coupled to the gauge wheel assembly. The rocker arm may be coupled to the float arm. The retainer may include a recess that releasably receives the pin to selectively lock the pin into the recess and a protrusion.

(6) The various aspects may include one or more of the following features. When the gauge wheel assembly is in the retracted position, the cam may engage the protrusion of the retainer to release the pin from the recess, placing the retainer in the unlocked configuration and the cutterbar in the flexible configuration. Extension of the gauge wheel assembly from the retracted position to the extended position may rotate the cam to release the retainer to cause the retainer to pivot and capture the pin into the recess, placing the retainer into the locked configuration and the cutterbar in the rigid configuration. A biasing force may pivot the retainer in response to release by the cam. The biasing force may be generated by a compressed spring. Retraction of the gauge wheel assembly from the extended position to the retracted position may rotate the cam to engage the protrusion of the retainer, causing the retainer to release the pin from the recess, placing the retainer in an unlocked configuration and the cutterbar in the flexible configuration. The actuation system may also include a spring coupled to the rocker arm. The spring may apply a moment to the rocker arm to pivot the rocker arm in a direction that moves the pin towards the recess of the retainer. The moment applied to the rocker arm by the spring may pivot the rocker arm towards the retainer when weight of the float arm is removed from the rocker arm. A retainer assembly may be included. The retainer assembly may include the retainer, a shaft pivotably coupled to the retainer, and a spring disposed on the shaft and configured to apply a moment to the retainer. The spring may compress in response to engagement of the protrusion of the retainer by the cam, and the compressed spring may pivot the retainer about a second axis when the cam disengages the protrusion of the retainer.

(7) Additionally, the various aspects may include one or more of the following features. The actuation system may include a rocker arm that includes a pin and pivotable about the first axis, a retainer, and a cam coupled to the gauge wheel. The rocker arm may be coupled to the float arm. The retainer may include a recess that releasably receives the pin to selectively lock the pin into the recess and a protrusion. The cam may be moveable with the gauge wheel assembly to selectively engage the protrusion of the retainer to move the retainer between a locked configuration and an unlocked configuration. Simultaneously one of moving the cutterbar into a flexible configuration in response to retraction of the gauge wheel and moving the cutterbar into a rigid configuration by an actuation system that is operated in response to retraction of the gauge wheel may include extending the gauge wheel from the retracted position to the extended position; pivoting the cam in response to extension of the gauge wheel to disengage the cam from the protrusion of the retainer; and receiving the pin into the recess in response to disengagement of the cam from the protrusion

of the retainer, thereby locking the cutterbar into the rigid configuration. The rocker arm may be pivoted in a direction that moves the pin towards the recess of the retainer. Pivoting the rocker arm may include applying a moment to the rocker arm with a spring. Pivoting the rocker arm in a direction that moves the pin towards the recess of the retainer may include removing weight of the float arm from the rocker arm so that a moment applied to the rocker arm is great enough to pivot the rocker arm in the direction that moves the pin towards the recess of the retainer. Simultaneously one of moving the cutterbar into a flexible configuration in response to retraction of the gauge wheel and moving the cutterbar into a rigid configuration by an actuation system that is operated in response to retraction of the gauge wheel may include retracting the gauge wheel from the extended position to the retracted position; pivoting the cam in response to retraction of the gauge wheel to engage the cam with the protrusion of the retainer; and releasing the pin from the recess in response to engagement of the cam with the protrusion of the retainer, thereby unlocking the cutterbar and placing the cutterbar into the flexible configuration.

(8) Also, the various aspects may include one or more of the following features. Extension of the gauge wheel assembly from the retracted position to the extended position may rotate the cam to release the retainer to cause the retainer to pivot and capture the pin into the recess, placing the retainer into the locked configuration and the cutterbar in the rigid configuration. Retraction of the gauge wheel assembly from the extended position to the retracted position may rotate the cam to engage the protrusion of the retainer, causing the retainer to release the pin from the recess, placing the retainer in an unlocked configuration and the cutterbar in the flexible configuration.

(9) Other features and aspects will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description of the drawings refers to the accompanying figures in which:

(2) FIG. 1 is an oblique view of an example draper header, according to some implementations of the present disclosure.

(3) FIG. 2 is oblique view of a portion of a frame of an example header, according to some implementations of the present disclosure.

(4) FIG. 3 a detail view of a portion of a frame of an example header, according to some implementations of the present disclosure.

(5) FIG. 4 is cross-sectional view of portions of an example lockout system, according to some implementation of the present disclosure.

(6) FIG. 5 is an oblique view of the lockout system of FIG. 4.

(7) FIG. 6 is a detailed, cross-sectional view of a portion of the lockout system of FIG. 4.

(8) FIG. 7 is another detailed, cross-sectional view of a portion of the lockout system of FIG. 4.

(9) FIG. 8 is a side view of a portion of an example header with an associated gauge wheel in a retracted position, according to some implementations of the present disclosure.

(10) FIG. 9 is a side view of the portion of the example header of FIG. 8 with the associated gauge wheel in an extended position.

(11) FIG. 10 is a detailed view of an example retainer assembly, according to some implementations of the present disclosure.

(12) FIG. 11 is a detailed view of another example actuation system, according to some implementation of the present disclosure.

(13) FIG. 12 is another detailed view of the actuation system of FIG. 11.

(14) FIG. 13 is a detailed cross-sectional side view of the example actuation system of FIG. 11 in which the actuation system is in a first configuration.

- (15) FIG. 14 is a detailed cross-sectional view of the example actuation system of FIG. 11 in which the actuation system is in a second configuration.
- (16) FIG. 15 is a detailed cross-sectional view of a portion of an example agricultural implement, according to some implementations of the present disclosure.
- (17) FIG. 16 is another detailed cross-sectional view of the portion of the agricultural implement of FIG. 15.

DETAILED DESCRIPTION

- (18) For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the implementations illustrated in the drawings, and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the disclosure is intended. Any alterations and further modifications to the described devices, systems, or methods and any further application of the principles of the present disclosure are fully contemplated as would normally occur to one skilled in the art to which the disclosure relates. In particular, it is fully contemplated that the features, components, and/or steps described with respect to one implementation may be combined with the features, components, and/or steps described with respect to other implementations of the present disclosure.
- (19) The present disclosure is directed to agricultural implements and, particularly, to draper headers that include automated lockout systems that are movable between a flexible configuration and a rigid configuration. Although various examples are provided in the context of draper headers, the scope of the present disclosure is applicable to other types of agricultural implements. In some implementations, the lockout systems are actuated in response to deployment of gauge wheels of the header. In the flexible configuration, float arms of the header are freely pivotable about respective axes, and, in the rigid configuration, the float arms are retracted into contact with a portion of the header, thereby providing a cutterbar attached to the float arms in a rigid configuration. Extension of the gauge wheels causes float arms and cutterbar to move into the rigid configuration. Retraction of the gauge wheels causes the float arms and cutterbar to move into the flexible configuration. In some instances, the lockout systems provide for abutting contact between the float arms and another portion of the frame without any adjustment provided either during manufacturing or in the field, thereby reducing manufacturing and maintenance time and complexity. Further, in some implementations, in the rigid configuration, torque applied to a lockout tube is reduced to zero or approximately zero, thereby reducing size, weight, and cost of the header. Additionally, lockout systems described herein are actuatable from a cab of a vehicle. Consequently, an operator avoids having to exit the cab of a vehicle in order to actuate a lockout system within the scope of the present disclosure.
- (20) Words of orientation, such as “up,” “down,” “top,” “bottom,” “above,” “below,” “leading,” “trailing,” “front,” “back,” “forward,” and “rearward” are used in the context of the illustrated examples as would be understood by one skilled in the art and are not intended to be limiting to the disclosure. For example, for a particular type of vehicle in a conventional configuration and orientation, one skilled in the art would understand these terms as the terms apply to the particular vehicle.
- (21) For example, as used herein, with respect to an endless belt, unless otherwise defined or limited, the term “forward” (and the like) corresponds to a forward direction of travel of the belt, with respect to a supporting frame (e.g., a frame of a draper header), during normal operation of the belt. Likewise, the term “rearward” (and the like) corresponds to a direction opposite the forward direction of travel of the belt. In this regard, for example, a “forward facing” feature on an endless belt may generally face in the direction that the belt travels during normal operation, while a “rearward facing” feature may generally face opposite that direction.
- (22) Also as used herein, with respect to a header (or components thereof), unless otherwise defined or limited, the term “leading” (and the like) indicates a direction of travel of the header during normal operation (e.g., the forward direction of travel of a harvester vehicle carrying a

header). Similarly, the term “trailing” (and the like) indicates a direction that is opposite the leading direction. In this regard, for example, a “leading” edge of a cutterbar of a draper header may be generally disposed at the front of the cutterbar, with respect to the direction travel of the draper header during normal operation (e.g., as carried by a harvester vehicle). Likewise, a “trailing” edge of the cutterbar may be generally disposed at the back or a side of the cutterbar opposite the leading edge, with respect to the direction of travel of the draper header during normal operation.

(23) FIG. 1 shows an example draper header **100** that includes a frame **102** that supports a first side conveyor **104**, a second side conveyor and **106**, and a center conveyor **108**. Each of the conveyors **104**, **106**, and **108** is configured as a belt-type conveyor extending over a respective circumferential length. The conveyors **104**, **106**, and **108** include endless belts **110**, **112**, and **114** that are moved in respective loops along the header **100** by motive devices, such as motors, gears, or internal belts. The conveyors **104** and **106** are disposed on opposing wings **116** and **118**, respectively, of the header **100**. In the illustrated example, the conveyor **104** includes two endless belts **110**, and the conveyor **106** and two endless belts **112**. In other implementations, the conveyors **104** and **106** may include additional or fewer endless belts. Further, although the conveyor **108** is shown as including a single endless belt **114**, in other implementations, the conveyor **108** may include additional endless belts. The endless belts **110**, **112**, and **114** are supported on two or more rollers of the respective conveyors **104**, **106**, and **108**. Although the draper header **100** is illustrated as a rigid or non-folding draper header, the scope of the present disclosure encompasses folding draper headers.

(24) In some implementations the endless belts **110**, **112**, and **114** may be formed from elastomer-impregnated fabric belts. Generally, the endless belts **110** and **112** may be rotated such that upper surfaces of the endless belts **110** and **112** move inward along the header **100** in respective directions **120** and **122**. In this way, material, such as severed crop material, is moved by the endless belts **110** and **112** to the center conveyor **108**, which, in turn, uses the endless belt **114** to move the material off of the header **100**. For example, the header **100** may offload the material onto a harvester vehicle to which the header **100** is attached. The header **100** also includes a cylindrical conveyor **124**. The cylindrical conveyor **124** receives severed crop material from the center conveyor **108** and carries the crop material rearward (i.e., in a direction **126**) through an aperture in the frame **102** located between the cylindrical conveyor **124** and the center conveyor **108** and, ultimately, into the harvester vehicle.

(25) In the illustrated example, various cleats **130** are fixed to the surface of each of the endless belts **110**, **112**, and **114**, with the cleats **130** generally extending in a direction transverse to the direction of travel of the respective endless belt **110**, **112**, or **114**, e.g., directions **120**, **122**, and **126**. In some implementations, the cleat **130** may extend less than an entire width of the endless belts **110**, **112**, and **114**. For example, one or more of the cleats **130** may extend only partially across the respective width of the endless belts **110**, **112**, and **114**, and, accordingly, may not extend to a leading edge or a trailing edge of the belts **110**, **112**, and **114**.

(26) The header **100** also includes a cutterbar **132** at a leading edge **133** of the header **100**. The cutterbar **132** cuts crop material, such as to sever crop material from a field. The cutterbar **132** extends laterally along the header **100**. In some implementations, the header **100** may also include a leading cover arranged between the cutterbar **132** and the endless belts **110**, **112**, and **114**, with the leading cover extending at least partly over a leading edge of at least the belts **110** and **112**. As shown in FIG. 1, for example, the header **100** includes an elongated row of interlocking crop ramp segments **134** that extend along the header **100**.

(27) As depicted, the header **100** also includes a trailing cover **136**, which may be in the form of a back sheet deflector or rear frame cover, separated from the crop ramp segments **134** by a width **138**. The trailing cover **136** serves to cover and protect various internal components of the header **100** and also generally defines a trailing end of a crop-carrying region of the header **100**. In some implementations, the cleats **130** extend over the entire width **138**. In other implementations, the cleats **130** extend over a portion of the width **138**.

(28) During a harvesting operation, a harvester vehicle carries the header **100** through an agricultural field in a nominal forward direction **140**. As the header **100** is moved across the field, the cutterbar **132** operates to sever the crops at a location adjacent to the ground. The severed crop material generally falls in a trailing direction (i.e., generally opposite the direction **140**), onto one or more of the three conveyors **104**, **106**, and **108**. The conveyor **104** on the wing **118** carries the crop material in the direction **120**, using the endless belts **110**, toward the center of the header **100**. The conveyor **106** carries the severed crop material in the direction **122**, using the endless belts **112**, toward the center conveyor **108**, and the center conveyor **108** carries the severed crop material in the direction **126** towards and underneath the cylindrical conveyor **124**. The severed crop material from the cylindrical conveyor **124** is transported in the direction **126** through the aperture in the frame **102** of the header **100** and into the agricultural harvester.

(29) In the illustrated example, the conveyor **104** and the conveyor **106** are similarly configured, although the conveyors **104** and **106** carry crop material in opposite directions **120** and **122**, respectively. In other implementations, the conveyors **104** and **106** can be configured differently. Generally, however, the description herein of the conveyor **104** is applicable to the conveyor **106**, as well as other conveyors of other implementations.

(30) FIG. 2 is a view of a portion of a frame **200** of a header **202**, which may be similar to the header **100**. The portion of the frame **200** illustrated corresponds to a portion of a wing **204** of the header **202**. The wing **204** may be similar to wing **118** of the header **100**. The frame **200** includes a beam **206** extending laterally along the frame **200**. A back section **208** is coupled to the beam **206** and extends therefrom. The back section **208** also extends laterally along the frame **200**. An outboard side section **207** connects to the beam **206** and the back section **208** and defines a lateral end of the frame **200**. A plurality of mounting brackets **210** are also coupled to the beam **206**. With the header **202** conventionally oriented, the brackets **210** general extend in a direction corresponding to a forward direction. The frame **200** also includes a laterally extending cross tube **212** that connects to each of the mounting brackets **210**. In some implementations, the cross tube **212** may have a square, rectangular, or circular cross-sectional shape and may define a central passage. However, the cross tube **212** may have other cross-sectional shapes. Float arms **214** are pivotably coupled to the mounting brackets **210**, and a cutterbar **216** is coupled to distal ends **218** of each of the float arms **214**. Similar to the cutterbar **132**, the cutterbar **216** is a reciprocating cutterbar.

(31) In some implementations, the mounting brackets **210** and corresponding float arms **214** are laterally separated from adjacent mounting brackets **210** and corresponding float arms **214** by approximately 2.5 feet (ft) (0.8 meters (m)). In other implementations, the lateral separation **215** may be greater than or less than 2.5 ft (0.8 m). In still other implementations, the lateral separation **215** may vary. Thus, in some implementations, the lateral separation **215** between some adjacent mounting brackets **210** and corresponding float arms **214** may be uniform while the lateral separation between other adjacent mounting brackets **210** and corresponding float arms **214** may be non-uniform.

(32) With the header **202** in an unsecured or flexible configuration, each of the float arms **214** are able to pivot independently of the other float arms **214**. As a result, when the float arms **214** are in contact with the ground and propelled over the ground, such as during a harvesting operation, each of the float arms **214** is able to follow a topography or contour of the ground. In response to the float arms **214** conforming movement to the contour of the ground, the cutterbar **216** flexes to also conform to the contour of the ground. As a result, a portion of the crop extending from the ground and remaining in a field may be generally consistent, e.g., a height by which the crop remaining in field extends from the ground is generally uniform.

(33) In a rigid configuration in which the float arms **214** are held in an abutting relationship against a portion of the frame **200**, such as the cross tube **212**, the float arms **214** are prevented from following a contour of the ground, and the cutterbar **216** is maintained in a generally straight and

rigid configuration, e.g., the cutterbar **216** maintains a generally straight, unbent shape.

(34) The header **202** also includes a lockout system **220** that is operable to move the float arms **214** and the cutterbar **216** between the flexible configuration and the rigid configuration. In some implementations, the header **202** includes a lockout system **220** for each wing **204**. The separate lockout systems **220** are operable to move the float arms **214** and associated portion of the cutterbar **216** of one wing between the rigid configuration and the flexible configuration independently of the float arms **214** and associated portion of the cutterbar **216** of the other wing. Thus, in some implementations, the header **202** may include two lockout systems **220**. In other implementations, the header may include a single lockout system **220** for all of the wings of the header **202**. In still other implementations, the header **200** may include more than two lockout systems.

(35) The lockout system **220** includes a rotatable component, which, in the example of FIG. 2, is a lockout tube **222**. In some implementations, the lockout tube **222** is in the form of a shaft. The lockout tube **222** extends laterally along the header **202** through apertures **224** formed in each of the mounting brackets **210**. The lockout tube **222** is rotatable relative to the mounting brackets **210** about a centerline **226**.

(36) The header also includes gauge wheels **228**, as shown in FIG. 3. In some implementations, the header **200** includes two gauge wheels **228** distributed laterally along each wing of the header **200**. In other implementations, the header **200** includes fewer or additional gauge wheels **228**. FIG. 3 illustrates a single gauge wheel **228**, although, as explained earlier, the scope of the disclosure is not so limited.

(37) A gauge wheel assembly **229** includes the gauge wheels **228** and an arm **230** to which the gauge wheels are rotatably coupled. The arms **230** are pivotably coupled to a mounting bracket **210**. The gauge wheels assemblies **229** are movable between a retracted position and an extended position. An actuation system **231** couples the gauge wheel assembly **229**, via arm **230**, to the lockout tube **222** such that extension or retraction of the gauge wheel, such as by rotation of the arm **230** about axis **234**, causes rotation of the lockout tube **222** about the centerline **226**. The axis **234** extends through a pivot where the arm **230** is coupled to the mounting bracket **210**. In the illustrated example, the actuation system **231** is a linkage (shown in more detail in FIG. 4) that extends between and couples the lockout tube **222** and the arm **230**.

(38) FIG. 3 is a detail view of a portion of a frame **300** of an example header **302** within the scope of the present disclosure. The frame **300** may be similar to the frame **200**. The frame **300** includes a laterally-extending beam **304**, similar to beam **206**, and a mounting bracket **306**, similar to mounting bracket **210**. A float arm **308** is pivotably mounted to the mounting bracket **306**. A gauge wheel **310** is coupled to the frame **300**. The gauge wheel **310** is rotatably coupled to an arm **312**, and the arm **312** is pivotably mounted to the mounting bracket **306**. A lockout tube **314**, similar to lockout tube **222**, extends laterally through apertures **316** formed in the mounting bracket **306**. The lockout tube **314** is rotatable about a centerline **318**, similar to centerline **226**, that extends longitudinally along the lockout tube **314**. The lockout tube **314** is rotatably coupled to the arm **312** via a linkage **320**. In the illustrated example, the linkage includes three links. However, in other implementations, the linkage **320** may include additional or fewer links. The linkage **320** includes a first link **322** fixedly attached to the arm **312**, a second link **324** pivotably coupled to the first link **322**, and a third link **325** fixedly attached to the lockout tube **314** and pivotably coupled to the second link **324**.

(39) The header **302** also includes a lockout system **326** that functions to move the float arms **308** and associated portions of a cutterbar provided at the distal ends of the float arms **308** of the header **302** between a flexible configuration and a rigid configuration. The lockout tube **314** is connected to the lockout system **326** such that rotation of the lockout tube **314** in a first direction causes the lockout system **326** to lockingly position the float arms **308** into the rigid configuration.

Conversely, rotation of the lockout tube **314** in a second direction, opposite the first direction, causes the lockout system **326** to move the float arms **308** from the rigid configuration into the

flexible configuration.

(40) Although some implementations include a lockout system, such as lockout system **326** described in more detail below, in other implementations, a lockout system may be omitted. Thus, in some implementations, the lockout **314** is coupled directly or indirectly to the float arms **308** without the use of a lockout system.

(41) FIG. **4** is a cross-sectional view of additional portions of the lockout system **326**. The lockout system **326** also includes a tensioner **400** and a linkage **402** coupled to the lockout tube **314**. The tensioner **400** includes a bracket **404**, a shaft **406** extending through an aperture **408** in a side **410** of the bracket **404**, and a biasing component **412** captured on the shaft **406** between the side **410** of the bracket **404** and a flange **414** secured to the shaft **406**. In some implementations, the flange **414** may be secured to the shaft between a shoulder **416** and a nut **418** threadably received onto the shaft **406**. In other implementations, the flange **414** may be secured to the shaft **406** in other ways, such as by welding, a press fit, or by being integrally formed onto the shaft **406**.

(42) In some implementations, the biasing component **412** is a spring, such as a coil spring. In some implementations, the biasing component **412** is a plurality of biasing components. For example, in some implementations, the biasing component **412** is a plurality of Bellville washers **413** stacked along a length of the shaft **406**, as shown in FIG. **4**. In some implementations, the Bellville washers are arranged in pairs, such that a base of each Bellville washer in a pair abuts each other. Pairs of the Bellville washers may be arranged adjacent to each other along a length of the shaft **406**, as shown, for example, in FIGS. **4**, **6**, and **7**. In some implementations, 32 Bellville washers may be used. However, additional or fewer Bellville washers may be used, and the number of Bellville washers may vary depending upon, for example, sizes and masses of the different components of a header.

(43) In still other implementations, the biasing component **412** may be or include a coil spring. For example, in some instances, the biasing component **412** may include a plurality of coils springs. One or more of the coils springs may be received onto the shaft **406**. In still other implementations, the biasing component **412** may be another type of spring.

(44) The tensioner **400** is pivotably coupled to the float arm **308** by a pin **420** coupled to the float arm **308**. In the illustrated example, the pin **420** extends through apertures **422** formed in a clevis **424** that is attached to the float arm **308**. The shaft **406** extends through a bore **425** formed through the pin **420**. A flange **426** captures the shaft **406** onto the pin **420**. In some implementations, the flange **426** may be a washer secured to the shaft **406** between a shoulder **428** and a nut **430** threadably received onto a threaded portion **432** of the shaft **406**. In other implementations, the flange **426** may be secured to the shaft **406** in other ways, such as a press fit or welding, or the flange **426** may be integrally formed on the shaft **406**. The shaft **406** also includes an enlarged portion **434** that abuts against the side **410** of the bracket **404**. Engagement between the side **410** and the enlarged portion **434** allows the biasing component **412** to be preloaded between the side **410** and the flange **414**. In some implementations, the biasing component **412** may not be preloaded.

(45) The preload applied to the biasing component **412** may be selected to ensure a force applied to the float arms **308** of a lockout system **326** by the biasing component **412** lifts the float arms **308** into abutting contact between all of the float arms **308** and a portion of the frame **300**, such as a cross tube similar to cross tube **212** shown in FIG. **2**. Thus, the preload ensures that a force ultimately provided by the biasing component **412** as the lockout system **326** is moved into the rigid configuration fully actuates all of the float arms **308** notwithstanding any variations in the header **302**, such as manufacturing variations that may otherwise prevent all of the float arms **308** from being in abutting contact with the cross tube when the lockout system **326** is in the rigid configuration. As a result, lockout systems of the present disclosure are operable to ensure full retraction of all of the float arms of a header when placed in the rigid configuration without adjustment during manufacturing or sometime later in the field, such as by a user or technician.

Thus, the lockout systems and associated headers of the present disclosure reduce maintenance thereto, improves performance of operation of the headers, increases productivity of the headers, and reduces costs of operation of the headers.

(46) The linkage **402** includes a first link **436** coupled to the lockout tube **314** and a second link **438** pivotably coupled to the first link **436** and the bracket **404**. In the illustrated example, the first link **436** is attached to the lockout tube **314** with a fastener **439**, such as a bolt. However, in other implementations, the first link **436** may be attached to the lockout tube **314** in other ways, such as by welding, interference fit, an adhesive, or by being integrally formed on the lockout tube **314**. Also, in the illustrated example, a nut **441** is used to secure the fastener **439** and the first link **436** to the lockout tube **314**.

(47) Referring to FIGS. 4 and 5, the bracket **404** has a general U-shape, and the second link **438** includes a first side **440** and a second side **442**. Free ends **444** of the bracket **404** are sandwiched between the first and second sides **440** and **442** at a first end **445** of the second link **438**. A tab **446** formed on the first link **436** is disposed between the first and second sides **440** and **442** of the second link **438** at a second end **452** of the second link **438**. A pin **448** extends through the first and second sides **440** and **442** at the second end **452** of the second link **438** and the tab **446** of the first link **436** to pivotably couple the first link **436** and the second link **438**. A pin **450** extends between the free ends **444** of the bracket **404** and the first and second sides **440** and **442** at the first end **445** of the second link **438** to pivotably couple the second link **438** and the bracket **404**. In some implementations, the pins **448** and **450** may be a rod or a fastener, such as a bolt. However, the pins **448** and **450** may have other forms to enable the first link **436** to pivot relative to the second link **438** and the bracket **404** to pivot relative to the second link **438**. The float arm **308** is pivotable about a pin **451** that pivotably couples the float arm **308** to the mounting bracket **306**. The pin **451** may be, for example, a fastener (e.g., a bolt), a shaft, or other component operable to permit pivoting movement of the float arm **308** relative to the mounting bracket **306**. FIG. 5 also shows an impact absorber component **460** that is attached to the frame **300** of the header **302**, such as the cross tube, and contacts a float arm **308** when retracted into the rigid configuration. The impact absorber component **460** may be attached to the cross tube with fasteners **462**, which may be, for example, bolts, pins, or rivets.

(48) As shown in FIGS. 4 through 7, the second link **438** has an arcuate shape that provides a relief or recess **437** that receives the lockout tube **314**. The recess **437** formed by the arcuate shape receives the lockout tube **314**, allowing the centerline **318** of the lockout tube **314** to intersect with centerline **454** of shaft **406**, resulting in the elimination of torque in the lockout tube **314**, as described in more detail below. In some instances, the centerlines **318** and **454** may be slightly offset due to slight variations in size of the components, movement of the different components, or variations in components, for example. These slight variations may produce an offset between the centerlines **318** and **454** that, in some cases, may be unavoidable. However, for the purposes of the present disclosure, intersection of the centerlines **318** and **454** is intended to encompass the slight offsets therebetween which may occur.

(49) As shown in FIG. 5, rotation of the lockout tube **314** in the direction of arrow **456** to a first position results in the float arms **308** being placed into a fully retracted position, which corresponds to the rigid configuration of the float arms **308** and cutterbar, such as cutterbar **216**. Rotation of the lockout tube **314** in the direction of arrow **458** to a second position results in the float arms **308** being placed in a fully extended position, which corresponds to the flexible configuration of the float arms **308** and cutterbar.

(50) FIGS. 4, 6, and 7 illustrate actuation of the lockout system **326** between the flexible configuration and the rigid configuration. In FIG. 6, the lockout system **326** is in the flexible configuration in which the floating arms **308** are in a fully extended. As a result, the float arms **308** are freely pivotably about pin **451**, and each of the float arms **308** coupled to the lockout system **326** are able to pivot independently of the other float arms **308**. Although the present example

describes a lockout system that can be included on a single wing of a header, in other implementations, a single lockout system operable to position all of the float arms of a header between the flexible configuration and a rigid configuration may be used.

(51) Referring to FIGS. 4 and 6, the lockout tube 314 is angularly oriented in the second position such that the biasing component 412 is unloaded, other than a preload that may be applied to the biasing component 412. With the lockout tube 314 in the second position, the float arms 308 are freely pivotable about the pin 451, allowing the float arms 308 to follow a contour of the ground when the float arms 308 are placed in contact with the ground. As the lockout tube 314 is rotated in the direction of arrow 456, the shaft 406 translates relative to and rotates with the pin 420. As a result, the shaft 406 is both rotated and translated towards the lockout tube 314. As shown in FIG. 6, the shaft 406 is displaced to cause the flange 426 to come into contact with the pin 420. Further rotation of the lockout tube 314 in the direction of arrow 456 results in further displacement and rotation of the shaft 406, which, in turn, causes further compression of the biasing component 412.

(52) With the flange 426 in contact with the pin 420, as the lockout tube 314 continues to be rotated in the direction of arrow 456, the float arm 308 is pivoted about the pin 451 in the direction of arrow 461. Moreover, as the shaft 406 is pivoted in the direction of arrow 460, an amount of torque applied to the lockout tube 314 decreases as the centerline 454 of the shaft 406 approaches the centerline 318 of the lockout tube 314.

(53) FIG. 7 shows the lockout system 326 in the rigid configuration. As shown in FIG. 7, the lockout tube 314 is moved into the first position. As the lockout tube 314 is moved from the position shown in FIG. 6 to the position shown in FIG. 7, the float arms 308 are retracted as a result of the contact between the flange 426 and the pin 420. With the locking system 326 in the rigid configuration, the float arms 308 are fully retracted and in abutting contact with the cross tube, such as cross-tube 212, or another component of the frame 300; the lockout tube 314 resides in the curved recess 437 formed by the second link 438; and the centerline 454 of the shaft 406 intersects the centerline 318 of the lockout tube 314. As a result of the intersection of the centerline 454 and the centerline 318, torque applied to the lockout tube 314 is reduced to approximately zero. Further, with the float arms 308 in the rigid configuration, the cutterbar is also placed into a straight and rigid configuration.

(54) With the torque applied to the lockout tube 314 being effectively zero when the float arms 308 are in the retracted and rigid configuration, a size of lockout tube 314 may be reduced, which results in a weight, size, and cost reduction. Additionally, compression of the biasing component 412 provides a force that is sufficient to retract all of the float arms 308 into abutting contact with a component of the frame 300, such as a cross-tube similar to cross-tube 212, notwithstanding any dimensional variations imparted to the frame 300 during manufacturing, for example.

Consequently, the lockout system 326 is operable to actuate all the float arms 308 into contact with a portion of the frame 300 without preliminary adjustment during manufacturing or subsequent adjustment when the header has entered use. Thus, the lockout system 326 avoids an adjustment performed during manufacturing or sometime thereafter, such as by a technician or user, to ensure full actuation of the float arms 308 into the rigid configuration.

(55) Automatic actuation of a lockout system, which may be similar to lockout system 326, in response to extension and retraction of a gauge wheel is described with reference to FIGS. 8 and 9. FIG. 8 is a side view of a portion of the example header 300 with the gauge wheel 310 in a retracted position, and FIG. 9 is a side view of the portion of the header 300 with the gauge wheel 310 in an extended position. The gauge wheel 310 and arm 312 define a gauge wheel assembly 800. The gauge wheel assembly 800 is pivotably coupled to the mounting bracket 306 at a pivot 802 and is rotatable about a pivot axis 804. The pivot 802 is defined by a shaft, bolt, or other component on which the gauge wheel assembly 800 rotates.

(56) The linkage 320 connects the gauge wheel assembly 800 and the lockout tube 314 and, hence, the lockout system 326. The second link 324 includes a pin 806 that is received into and a slot 808

defined by the third link **325**. The pin **806** is slidable within the slot **808**. In other implementations, the pin **806** and slot **808** are omitted, and the second link **324** and third link **325** are pivotably connected.

(57) When the gauge wheel assembly **800** is deployed from the stowed or retracted position, such as by an input made by an operator of a header **300** or agricultural vehicle to which the header **300** is coupled, the gauge wheel assembly **800** rotates in a direction of arrow **801** about the pivot axis **804**. In some instances, an actuator is used to position the gauge wheel assembly **800** between the retracted position and the extended position. In some implementations, the actuator is a hydraulic cylinder, a pneumatic cylinder, an electrically operated linear actuator, a rotary actuator, or some other actuator that is operable to move one or more gauge wheel assemblies between a retracted position and an extended position. In response, the second link **324** pivots relative to the first link **322** at pivot **810** while the pin **806** of the second link **324** slides within the slot **808**. As a result of the relative, sliding movement of the second link **324** and the third link **325** coupled to the lockout tube **314**, the lockout system **326** remains in an unaltered condition. In some instances, the third bracket **325** is fixedly attached to the lockout tube **314**, such as with a fastener (e.g., a bolt) or welding, friction fit, keyed engagement, spline engagement, or by being integrally formed with the lockout tube **314**. Thus, in the illustrated implementation, the lockout system **326** remains in an unlocked condition, resulting in the float arms **308** being in an extended or flexible condition, during a portion of the deployment of the gauge wheel assembly **800**. When the pin **806** reaches a first end **812** of the slot **808**, the second link **324** engages the third link **325**, causing the third link **325** and lockout tube **314** to rotate about centerline **318** in the direction of arrow **814**.

(58) In some implementations, the second link **324** and the third link **325** engage each other (i.e., when the pin **806** of the second link **324** engages the first end **812** of the slot **808** of the third link **325**) at selected amount articulation of the gauge wheel assembly **800** between the retracted position and the extended position. For example, in some implementations, the second link **324** and the third link **325** engage at an amount of articulation that results in the float arms **308** and the cutterbar of the header **302** being in the rigid configuration and 90 percent of articulation of the gauge wheel assembly **800**. In other implementations, the selected amount of articulation may be greater than or less than 90 percent. In some implementations, the second link **324** engages the third link **325** when the gauge wheel assembly **800** reaches a selected amount of articulation. For example, in some implementations, the second link **324** and the third link **325** engage each other when the gauge wheel assembly **800** reaches 90 percent of articulation from the retracted position to the extended position. Consequently, in such examples, over the remaining 10 percent of articulation, the lockout system **326** moves the float arms **308** and cutterbar of the header **302** from the flexible configuration to the rigid configuration. Again, in other implementations, the selected amount of articulation at which the second link **324** engages the first link **325** may be greater than or less than 90 percent.

(59) As mentioned above, when the second link **324** and the third link **325** engage, the third link **325** rotates in the direction of arrow **814**, rotating the lockout tube **314** also in the direction of arrow **814**. As the lockout tube **314** approaches full rotation (at which point the float arms **308** are in the rigid configuration), a pin **816** included on a bracket **818** coupled to the lockout tube **314** engages a retainer assembly **820**. In some instances, the bracket **818** forms part of the third link **325**. The retainer assembly **820** includes a retainer **822** that functions to capture and retain the bracket **818**, locking the lockout tube **314** and maintaining the associated float arms and cutterbar in the rigid configuration, as shown in FIG. 9.

(60) In the illustrated example of FIGS. 8 and 9, the float arms and cutterbar are locked in the rigid configuration within the final ten percent of extension travel of the gauge wheel assembly **800**. That is, a range of motion of the gauge wheel assembly **100** is defined between the retracted position and the extended position. As the gauge wheel assembly **800** reaches a position at approximately 90 percent of travel of the range of motion from the retracted position to the

extended position, the float arms and cutterbar begin to transition between the flexible configuration and the rigid configuration. By the time the gauge wheel assembly **800** reaches a position corresponding to 100 percent of travel of the range of motion, the float arms and cutterbar are locked into the rigid configuration.

(61) In other implementations, a location along the range of travel of the gauge wheel assembly **800** where the float arms and cutterbar begins to transition from the flexible configuration to the rigid configuration may be any desired position. For example, in some instances, transition of the float arms and cutterbar from the flexible configuration to the rigid configuration is completed within the first 15 percent of the range of motion of the gauge wheel assembly **800**. In other instances, the transition from the flexible configuration to the rigid configuration may begin at other points along the range of travel, such as 10 percent, 20 percent, 25 percent, 30 percent, or 40 percent of the range of travel. In still other implementations, the transition from the flexible configuration to the rigid configuration may begin at other positions. Similarly, the point along the range of movement at which the float arms and cutterbar reach the rigid configuration may be any selected point. Further, a size of a portion of the range of movement over which the float arms and cutterbar transition from the flexible configuration to the rigid configuration may be any desired amount. For example, the transition may occur over five percent, 10 percent, or 15 percent of the range of motion of the gauge wheel assembly from the retracted position to the extended position. In other implementation, a different amount of the range of motion may be used.

(62) Still further, although numerous examples are provided in the context of moving the float arms and cutterbar into the rigid configuration from the flexible configuration, this description is also applicable to movement of the float arms and cutterbar between the rigid configuration to the flexible configuration. For example, the float arms and cutterbar may begin to transition at the 90 percent point of the range of motion of the gauge wheel assembly **800** from the extended position to the retracted position. In other implementations, the transition may occur at other points along the range of motion, such as at 10 percent, 20 percent, 25 percent, 30 percent, or 40 percent of the range of travel. However, in other instances, the transition may initiate at other points along the range of motion. Further, in some implementations, the same point along the range of motion is used as the point where the transition occurs from the flexible configuration to the rigid configuration and vice versa.

(63) Referring to FIGS. **9** and **10**, as the float arms and cutterbar are placed into the rigid configuration, the pin **816** provided on the bracket **818** is received into a recess **900** formed in the retainer **822** of the retainer assembly **820**. A portion **902** of the retainer **822** forming a part of the recess **900** extends partially around the pin **816**. That is, in some implementations, the recess **900** extends more than 180 degrees around a perimeter of the pin **816** so that the pin **816** remains retained within the recess **900**, as described in more detail below.

(64) The retainer assembly **820** includes the retainer **822** that defines the recess **900**, a bracket **902** pivotably coupled to the retainer **822** at a pivot **904**, a rod **906** extending from the bracket **902**, and a spring **908** retained on the rod **906**. The rod **906** is coupled to a bracket **910**, and the spring **908** is captured between the brackets **902** and **910**. In some implementations, the spring **908** is preloaded (e.g., stretched) between the brackets **902** and **910**, causing the spring **908** to impart a pulling force on the bracket **902** that pulls the bracket **902** in the direction of arrow **912**. The retainer **822** is pivotably coupled to the mounting bracket **306** at a pivot **914**.

(65) The rod **906** includes an elongated portion **907** and an enlarged portion **909**. The elongated portion **907** extends through an opening **916** formed in a flange **917** of the bracket **902**. The enlarged portion **909** abuts the flange **917** of the bracket **902**. The rod **906** is retained on the bracket **902** by engagement between the enlarged portion **909** of the rod **906** and the flange **917** of the bracket **902**, and the rod **906** is retained on the bracket **910**, such as by a fastener **918**. For example, in some instances, an end of the rod **906** is threaded and threadingly mates with a fastener **920**, such as a nut. Consequently, the rod **906** is captured between brackets **902** and **910**. In some

implementations, the rod **906** and the bracket **902** form an integral component.

(66) The spring **908** is secured to the bracket **902** and the bracket **910** such that the spring **908** is stretched between the bracket **902** and bracket **910**. For example, in some implementations, the spring **908** engages the enlarged portion **909** of the rod **906**, such as an end surface **919** of enlarged portion. Further, the elongated portion **907** of the rod **906** extends through and is slidable within an opening **921** formed in the bracket **910**.

(67) As the pin **816** is received into the recess **900**, the pin **816** engages to retainer **822**, causing the retainer **822** to pivot about the pivot **914** a few degrees in the direction of arrow **924**. Because the pivots **904** and **914** are offset from each other by an amount D relative to a line **922** that is parallel to a direction of force applied by the spring **908** (which, in this example, passes through a centerline of the rod **906** and a center of the pivot **904**), causes further expansion of the spring **908** (i.e., further stretching of the spring **908**). As a result, a force generated by the spring **908** increases, generating a moment that biases the retainer **822** to move in a direction of arrow **926** about the pivot **914**. When the pin **816** is fully received into the recess, the retainer **822** rotates back in the direction of arrow **926** in response to the generated moment and captures the pin **816** in the recess **900**. The shape of the recess **900** in combination with the moment created by the biasing force produced by the spring **908** captures and prevents the pin **816** from being removed from the recess **900**. Consequently, the retainer assembly **820** captures the pin **816** and the moment generated by the spring **908** (i.e., the biasing of the retainer **822** in the direction of arrow **926** about the pivot **914**) maintains the float arms and cutterbar in the rigid configuration.

(68) In other implementations, the spring **908** is placed in a compressed state, as opposed to a stretched state. For example, relocating the pivot **914** to a location between the recess **900** and the line **922** causes the spring **908** to be compressed in response to rotation of the retainer **822** in the direction of arrow **924**. In such instances, the spring **908** may be preloaded by placing the spring **908** in compression. Thus, the spring **908** may be placed in tension (i.e., stretched) or compression to provide the locking functionality of the retainer assembly **820**, such as by altering a position of the pivot **914** relative to the recess **900** and the line **922**.

(69) With continued reference to FIGS. **9** and **10**, as the gauge wheel assembly **800** is retraced (i.e., by the rotation of the gauge wheel assembly **800** in the direction of arrow **803** about pivot axis **804** from the extended position to the retracted position), the pin **806** slides within the slot **808** as the second link **324** moves relative to the third link **325**. As the gauge wheel assembly **800** continues to rotate in the direction of arrow **803**, the pin **806** reaches and engages the second end **807** of the slot **808**, causing engagement of the second link **324** and the third link **325**. In response, the third link **325** and bracket **818** rotate in the direction of arrow **805** about centerline **318**. In turn, the pin **816** overcomes retention by the retainer **822** and is removed from the recess **900**, and the float arms and cutterbar are released from the rigid configuration. Removal of the pin **816** from the recess **900** cause a slight rotation of the retainer **822** about pivot **914** in the direction of arrow **924**. The biasing force generated by the spring **908** in response causes the retainer **822** to rotate in the direction of arrow **926** upon release of the pin **816**. As a result, the float arms and cutterbar are moved from the rigid configuration to the flexible configuration.

(70) FIGS. **11-14** illustrate another example actuation system **1100** for actuating a lockout system that may be similar to the lockout system **326**, described earlier. The actuation system **1100** includes a lever **1102** coupled to a gauge wheel assembly **1104** (which may be similar to the gauge wheel assembly **229** or gauge wheel assembly **800**), a rocker arm **1106** coupled to a lockout tube **1108** (which may be similar to lockout tube **314**), and a retainer **1110** that is pivotably coupled to a mounting bracket **1112** (which may be similar to the mounting bracket **306**). The lockout tube **1108** is coupled to the lockout system in manner that may be similar to the manner described above with respect to lockout tube **314** and lockout system **326**. Rotation of the lockout tube **1108** operates to actuate the lockout system to move one or more float arms (which may be similar to float arms **308**) and a cutterbar (which may be similar to cutterbar **132**) between a rigid configuration and a

flexible configuration, as described earlier.

(71) In the illustrated example, the rocker arm **1106** includes a semicircular portion **1114** that abuts the lockout tube **1108**, and the rocker arm **1106** is secured to the lockout tube **1108** using fasteners **1116** that extend through the lockout tube **1108** and the semicircular portion **1114**. In other implementations, the rocker arm **1106** and the lockout tube **1108** can be joined in other ways, such as by welding or by using an adhesive or by integrally forming the rocker arm **1106** with the lockout tube **1108**. As shown, the rocker arm **1106** includes first and second sides **1118** and **1120**, laterally offset from each other, with first and second pins **1122** and **1124** extending therebetween. The first and second pins **1122** and **1124** engage respective recesses **1126** and **1128** formed in the lever **1102** and retainer **1110**, as described in more detail below.

(72) The retainer **1110** is pivotable on the mounting bracket **1112** about a pivot **1130**. In the illustrated example, the retainer **1110** is pivotably mounted the mounting bracket **1112** via bracket **1132**. During actuation of the actuation system **1100**, the lever **1102** and the retainer **1110** are receivable between the first and second sides **1118** and **1120** of the rocker arm **1106**.

(73) The retainer **1110** forms part of a retainer assembly **1134**. The retainer assembly **1134** includes the retainer **1110**, a bracket **1136** pivotably coupled to the retainer **1110** at a pivot **1138**, a rod **1140** captured between the bracket **1136** and a bracket **1142** attached to the mounting bracket **1112**, and a spring **1144** retained on the rod **1140** between the bracket **1136** and the bracket **1142**. The rod **1140** includes an elongated portion **1146** and an enlarged portion **1148**. The elongated portion **1146** extends through an opening **1150** formed in a flange **1152** of the bracket **1136** and through an opening **1154** formed in the bracket **1142**. The rod **1140** is retained on the bracket **1142** by a fastener **1156**. For example, in some instances, an end of the rod **1140** is threaded and threadingly mates with the fastener **1156**, which may be a nut. Consequently, the rod **1140** is captured between brackets **1136** and **1142**. A position of the fastener **1156** along the elongated portion **1146** of the rod **1140** defines how far the bracket **1136** is able to move away from the bracket **1142** and, thus, how far the spring **1144** is able to expand. In some implementations, the rod **1140** and the bracket **1136** are integrally formed.

(74) In some implementations, the spring **1144** is preloaded (e.g., compressed) between the brackets **1136** and **1142**, causing the spring **1144** to impart a pushing force on the bracket **1136** that pushes the bracket **1136** in the direction of arrow **1158** (parallel to centerline **1160** of the rod **1140**), tending to rotate retainer **1110** in the direction of arrow **1162** about pivot **1130**. The retainer **1110** is pivotably coupled to the mounting bracket **1112** at a pivot **1130**.

(75) Operation of the actuation system **1100** is described with reference to FIGS. **13** and **14**. In operation, with the gauge wheel assembly **1104** in the retracted position, extension of the gauge wheel assembly **1104** is selected, such as by an operator of an agricultural implement that includes the gauge wheel assembly **1104**. In response, the gauge wheel assembly **1104** is rotated about a pivot axis **1162** of pivot **1164** in the direction of arrow **1166**. As the lever **1102** is rotated with the gauge wheel assembly **1104**, the interaction between the recess **1126** and the first pin **1122** causes the rocker arm **1106** to rotate in the direction of arrow **1168** about a centerline **1170** of the lockout tube **1108**. Being coupled with the rocker arm **1106**, the lockout tube **1108** similarly rotates about the centerline **1170** in the direction of arrow **1168**, moving the float arms and cutterbar from the flexible configuration to the rigid configuration.

(76) As the gauge wheel assembly **1104** continues to move into the extended position, the lever **1102** continues to rotate the rocker arm **1106** in the direction of arrow **1168**, causing the second pin **1124** to engage the retainer **1110** and be received within the second recess **1128**. In some instances, prior to the second pin **1124** being inserted into the second recess **1128**, the second pin **1124** contacts an outer edge of the retainer **1110**, causing the retainer **1110** to rotate a few degrees in the direction of arrow **1172**. Thereafter, the second pin **1124** enters the second recess **1124**, and the retainer **1110** rotates a few degrees back in the direction of arrow **1162** in response to a spring force generated by the spring **1144**.

(77) With the second pin **1124** received within the second recess **1128**, the second pin **1124** causes the retainer **1110** to rotate in the direction of arrow **1172** about pivot **1130**. With the second pin full received into the second recess **1128** and the retainer **1110** fully rotated in response, continued rotation of the gauge wheel assembly **1104** cause the first pin **1122** to be withdrawn from the recess **1126**. As a result, rotation of the rocker arm **1106** ceases. At this point, the lockout tube **1108** is fully actuated to cause the float arms and cutterbar to be locked into the rigid configuration.

(78) Further, as a result of rotation of the retainer **1110** in the direction of arrow **1172**, the retainer **1110** displaces the bracket **1136** in the direction of arrow **1174**, parallel to the centerline **1160** of the rod **1140**, compressing the spring **1144**. Compression of the spring **1144** (along with any preload applied to the spring **1144**) produces a biasing force in the direction of arrow **1158**. A shape of the recess **1128** interacts with the pin **1124** to prevent withdrawal of the pin **1124** from the recess **1128**, and the produced biasing force generated by the spring **1144**, in combination with the interaction between the pin **1124** and the recess **1128**, lock the rocker arm **1106** and the retainer assembly **1134** together. FIG. **14** shows a condition of the actuation system **1100** when the gauge wheel assembly **1104** is in the extended position.

(79) To retract the gauge wheel assembly **1104**, an input is provided, such as by the operator of the implement, to retract the gauge wheel assembly **1104**. In response, the gauge wheel assembly **1104** is rotated in the direction of arrow **1176**. As the gauge wheel assembly **1104** rotates in the direction of arrow **1176**, the lever **1102** engages the first pin **1122**, causing the first pin **1122** to be received into the first recess **1126**. Particularly, a finger **1135** extending along a side of the first recess **1126** engages the first pin **1122** and directs the first pin **1122** into the first recess **1126** as the lever **1102** is rotated in the direction of arrow **1176**. With further rotation of the gauge wheel assembly **1104** in the direction of arrow **1176**, engagement between the first pin **1122** and the lever **1102** causes rocker arm **1106** and the lockout tube **1108** to rotate in the direction of arrow **1178**, which, in turn, causes the second pin **1124** to be removed from the second recess **1128**. At this point, the lockout system unlocks, placing the cutterbar and float arms in the flexible configuration. In some instances, removal of the second pin **1124** from the second recess **1128** cause the retainer **1110** to be rotated a few degrees in the direction of arrow **1172**. This rotation of the retainer **1110** causes an increased compression of the spring **1144**. The biasing force generated by the second spring **1144** produces a moment that rotates the retainer **1110** back to an initial position. FIG. **13** illustrates the gauge wheel assembly **1104** in the fully retracted position.

(80) In this way, the actuation system **1100** is operable to move the cutterbar and float arms into and between the rigid configuration and the flexible configuration in response to movement of the gauge wheel assembly **1104**. Although a single gauge wheel assembly **1104** is illustrated, an implement within the scope of the present disclosure may include more than one gauge wheel assemblies that operate in a similar manner.

(81) Similar to the spring **908**, the spring **1144** can be configured to operate in a stretched or compressed condition by, for example, altering a position of the pivot **1130** relative to recess **1128** and longitudinal axis **1160**.

(82) FIGS. **15** and **16** illustrate another example actuation system **1500** that is operable to move a cutterbar, which may be similar to the cutterbar **132** (shown in FIG. **1**), and associated float arms, which may be similar to float arms **214** (shown in FIG. **2**), between a locked, rigid configuration and an unlocked, flexible configuration in response to extension and retraction of a gauge wheel assembly **1502**, which may be similar to gauge wheel assembly **1104**. Although a single gauge wheel assembly **1502** is illustrated, in other implementations, additional gauge wheel assemblies **1502** may be included.

(83) Referring to FIG. **15**, the actuation system **1500** includes a cam **1504**, a rocker arm **1506**, a retainer assembly **1508**, and a spring **1510**. In some implementations, the cam **1504** is in the form of a plate that is coupled to an arm **1512** of the gauge wheel assembly **1502**. The gauge wheel assembly **1502** includes the arm **1512** and a wheel pivotably coupled to the arm **1512**. The wheel

may be similar to the wheel **310**, shown in FIG. **3**, discussed earlier. The gauge wheel assembly **1502** is pivotably coupled to a mounting bracket **1516**, which may be similar to mounting bracket **1112**, described earlier. The rocker arm **1506** is fixedly attached to a lockout tube **1518**, which may be similar to lockout tube **1108**, described earlier. In some implementations, the rocker arm **1506** has a construction similar to that of the rocker arm **1106**. That is, in some instances, the rocker arm **1506** includes first and second sides, which may be similar to the first and second sides **1118** and **1120** of rocker arm **1106**, and pin **1519** extends between the first and second sides. Thus, when the pin **1519** is received in recess **1544**, as described below, the retainer **1520** is received between the sides of the rocker arm **1506**.

(84) The rocker arm **1506** rotates with the lockout tube **1518** relative to the mounting bracket **1516**. Further, the rocker arm **1506** may be attached to the lockout tube **1518** in a variety of ways, such as those described above with respect to lockout tube **1108** and rocker arm **1106**. The rocker arm **1506** includes the pin **1519** disposed on a first end **1523** of the rocker arm **1506**.

(85) The retainer assembly **1508** has a construction that is similar to that of the retainer assembly **1134**. The retainer assembly **1508** includes a retainer **1520** that is pivotable coupled to the mounting bracket **1516** via a pivot **1521**, a bracket **1522** pivotably coupled to the retainer **1520** at a pivot **1524**, a rod **1526** extending between the bracket **1522** and a bracket **1528** attached to the mounting bracket **1516**, and a spring **1530** captured on the rod **1526** between the bracket **1522** and the bracket **1528**. The rod **1526** extends through and is movable in an opening **1532** formed in the bracket **1528**, such as in the manner described above in the context of rod **1140**. Thus, the rod **1526** is movable in a direction parallel to a longitudinal axis **1534** of the rod **1526** relative to the bracket **1528**, and an amount by which the spring **1520** is able to expand is limited by a fastener **1525** or another component coupled to the rod **1526**, thereby limiting an amount by which the rod is movable in the opening **1532**. Thus, in some implementations, the rod **1526** is retained to the bracket **1528** in a manner similar to that described above with respect to rod **1140** and bracket **1142**. In the illustrated example, the spring **1510** is connected to a beam **1536** (which may be similar to beam **304**) at a first end **1538** via a bracket **1540** and, at a second end **1541**, to a second end **1542** of the rocker arm **1506**. In other implementations, the spring **1510** may be coupled to another part of the header, such as another part of the header frame, e.g., the mounting bracket **1516**. The retainer **1520** includes the recess **1544** that removably receives the pin **1519** of the rocker arm **1506**.

(86) The cam **1504** includes a cam shape that engages with a counterpart protrusion **1546** formed on the retainer **1520**. As the cam **1504** rotates with the gauge wheel assembly **1502**, the cam shape of the cam **1504** and the protrusion **1546** of the retainer **1520** operate to rotate the retainer about pivot **1521**.

(87) FIG. **15** shows the gauge wheel assembly **1502** in the retracted position in which the float arms and cutterbar are in the flexible configuration, and FIG. **16** shows the gauge wheel assembly **1502** in the extended position with the actuation system **1500** configured such that the lockout tube **1518** has locked the float arms and cutterbar in the rigid configuration. In normal operation, when the gauge wheel assembly **1502** is in the retracted position, the float arms of the agricultural implement are in contact with the ground. In the retracted position, the cam **1504** is in contact with the protrusion **1546** of the retainer **1520**, maintaining the retainer **1520** in an unlocked configuration. In the unlocked configuration, the float arms and cutterbar, via the float arms, are in the flexible configuration and are free to pivot about axis **1548**. As shown in FIG. **5**, this pivot axis is defined by one or more pins, such as pin **451** in FIG. **5**.

(88) Generally, the lockout tube **1518** is connected to a plurality of float arms. When the cutterbar and float arms are in the flexible configuration and the associated implement is moved through a field, such as during a harvesting operation, the float arms are allowed to follow a contour of the ground, thus, causing the float arms to pivot about axis **1548**. If at least one of the float arms pivots downwards, i.e., in the direction of arrow **1549**, the float arm causes the lockout tube **1518** and

rocker arm **1506** to rotate in the direction of arrow **1551** as a result of a lockout system similar to lockout system **326** coupling the float arm to the lockout tube **1518**. This position of the lockout tube **1518** and rocker arm **1506** is shown in FIG. **15**, and this position is the prevailing one during an agricultural operation, such as harvesting. This is because, generally, at least one float arm is at least partially in a downwardly pivoted position, causing the lockout tube **1518** to rotate in the direction of arrow **1551**. In some implementations, the spring **1510** provides enough force to rotate the lockout tube **1518** and associated rocker arms **1506** about axis **1552** in the direction of arrow **1550** (such as to a position similar to that shown in FIG. **16**) when all of the weight of the float arms is removed from the lockout tube **1518**. Consequently, when moving the float arms and cutterbar from the flexible configuration to the rigid configuration, all of the float arms are fully pivoted in the direction of arrow **1553**, such as until each of the float arms contacts a stop, thereby removing all of the weight of the float arms and cutterbar from the lockout tube **1518**. In response, a force exerted by the spring **1510** to the lockout tube **1518** (via the rocker arms **1506**) causes the lockout tube **1518** and associated rocker arms **1506** to rotate in the direction of arrow **1550** and attain a position similar to that shown in FIG. **16**. At this location, the float arms and cutterbar can be locked into the rigid configuration, as described in more detail below.

(89) With the weight of the float arms and cutterbar being supported by the ground and as a result of the rotation to the lockout tube **1518** and the rocker arm **1506** due to the spring **1510**, the lockout tube **1518** is maintained in a position similar to that of the lockout tube **314** shown in FIG. **7**. That is, the lockout tube **1518** is maintained in a position that is or is close to a locked position, as shown, for example, in FIG. **7** (referred to as the first position in the description associated with FIG. **7**). This rotation of the lockout tube **1518** and rocker arm **1506** cause the pin **1519** to be received into the recess **1544** formed in the retainer **1520**.

(90) When the gauge wheel assembly **1502** is moved from the retracted position to the extended position, such as by an actuator, the gauge wheel assembly **1502** rotates about pivot **1554** in a direction of arrow **1556**. As the rotation of the gauge wheel assembly **1502** occurs, the cam **1504** disengages from the protrusion **1546** of the retainer **1520**, which, in return, causes the compressed spring **1530** to expand and exert a moment on the retainer **1520**. The retainer **1520** rotates about pivot **1521** in the direction of arrow **1531** in response to the moment, locking the rocker arm **1506** and the lockout tube **1518** into position. Further, locking of the lockout tube **1518** and the rocker arm **1506** into position by the retainer **1520** occurs before all of the weight of the float arms and cutterbar is removed from the ground. Consequently, the float arms and cutterbar are locked into position during an initial range of rotation during extension of the gauge wheel assembly **1502**. As rotation of the gauge wheel assembly **1502** continues to the point where all of the weight of the float arms and cutterbar is removed from the ground and is, therefore, no longer supported by the ground, the float arms and cutterbar are locked into the rigid configuration. FIG. **16** shows the actuation system **1500** in a configuration in which the float arms and cutterbar are locked in the rigid configuration.

(91) Thus, the actuation system **1500** is operable to lock the float arms and cutterbar of an implement into a rigid configuration within a portion of the actuation of a gauge wheel assembly during movement from a retracted position to an extended position. In some implementations, locking the float arms and cutterbar into the rigid configuration occurs within 15 percent of the range of movement between the retracted position and the extended position. In other implementations, locking occurs at other percentages of movement. For example, locking may occur at five percent, ten percent, 20 percent, 25 percent, or 30 percent. In still other implementations, locking may occur at percentages of the movement of the gauge wheel assembly that is less than five percent, greater than 30 percent, or at some point between the specific percentages described.

(92) The gauge wheel assembly **1502** is retracted into the retracted position to move the cutterbar and float arms from the rigid configuration into the flexible configuration. Referring again to FIGS.

15 and 16, with the gauge wheel assembly 1502 in the extended position (shown in FIG. 16), the gauge wheel assembly 1502 is retracted, such as in response to an operator input. During retraction, the gauge wheel assembly 1502 is rotated about the pivot 1554 in the direction of arrow 1558. During rotation of the gauge wheel assembly 1502, the cam 1504 engages the protrusion 1546 of the retainer 1520, causing the retainer 1520 to rotate in the direction of arrow 1560. This releases the pin 1519 from the recess 1544, thereby unlocking the lockout tube 1518 and rocker arm 1506 and placing the float arms and cutterbar into the flexible configuration. The gauge wheel assembly 1502 continues to rotate about the pivot 1554 in the direction of arrow 1558 until the gauge wheel assembly 1502 reaches the retracted position, as shown in FIG. 15.

(93) Locking the cutterbar into the rigid configuration during a selected amount of rotation of the gauge wheel assembly as the gauge wheel assembly is moved from the retracted position to the extended position is important because, in some instances, the gauge wheels may not be fully deployed to the extended position. This may be due, for example, to user preference or ground topography. In any case, since the gauge wheels can be extended to any number of positions between the retracted position and the extended position, it is desirable to have the cutterbar locked into the rigid configuration for any of those gauge wheel assembly deployed positions. Therefore, the actuation system 1500 is operable to place the float arms and cutterbar into a rigid configuration after a small amount of extension of the gauge wheel assembly 1502 and, in some case, almost immediate locking once the gauge wheel assembly 1502 has started to deploy away from the retracted position. Consequently, the actuation system 1500 provides for providing the float arms and cutterbar in the rigid configuration for a range of deployments of the gauge wheel assembly 1502 that is less than the extended position, which, in such cases may be referred to as a fully extended position.

(94) Without in any way limiting the scope, interpretation, or application of the claims appearing below, a technical effect of one or more of the example implementations disclosed herein is to automatically move a cutterbar of an agricultural implement between a rigid configuration and a flexible configuration in response to extension or retraction of a gauge wheel (e.g., a gauge wheel assembly). Another technical effect of one or more of the example implementations disclosed herein is to provide for locking a cutterbar into a rigid configuration within a portion of a total amount of articulation of a gauge wheel during deployment of the gauge wheel into an extended position. This provides for placing the cutterbar into a rigid configuration for different amounts of deployment of the gauge wheel or gauge wheels.

(95) While the above describes example implementations of the present disclosure, these descriptions should not be viewed in a limiting sense. Rather, other variations and modifications may be made without departing from the scope and spirit of the present disclosure as defined in the appended claims.

Claims

1. A system for automatically configuring a cutterbar of an agricultural implement comprising: a cutterbar; a float arm engaged with the cutterbar; a gauge wheel assembly; and an actuation system that connects the float arm and the gauge wheel, the float arm movable, via the actuation system, in response to retraction or extension of the gauge wheel to move the cutterbar between a rigid configuration and a flexible configuration, the actuation system comprising: a rocker arm comprising a pin and pivotable about a first axis, the rocker arm coupled to the float arm; a retainer comprising: a recess that releasably receives the pin to selectively lock the pin into the recess; and a protrusion; and a cam coupled to the gauge wheel assembly, the cam moveable with the gauge wheel assembly to selectively engage the protrusion of the retainer to move the retainer between a locked configuration and an unlocked configuration.

2. The system of claim 1, wherein, when the gauge wheel assembly is in the retracted position, the

cam engages the protrusion of the retainer to release the pin from the recess, placing the retainer in the unlocked configuration and the cutterbar in the flexible configuration.

3. The system of claim 1, wherein extension of the gauge wheel assembly from the retracted position to the extended position rotates the cam to release the retainer to cause the retainer to pivot and capture the pin into the recess, placing the retainer into the locked configuration and the cutterbar in the rigid configuration.

4. The system of claim 3, wherein a biasing force pivots the retainer in response to release by the cam.

5. The system of claim 4, wherein the biasing force is generated by a compressed spring.

6. The system of claim 1, wherein retraction of the gauge wheel assembly from the extended position to the retracted position rotates the cam to engage the protrusion of the retainer, causing the retainer to release the pin from the recess, placing the retainer in an unlocked configuration and the cutterbar in the flexible configuration.

7. The system of claim 1, wherein the actuation system further comprises a spring coupled to the rocker arm, wherein the spring applies a moment to the rocker arm to pivot the rocker arm in a direction that moves the pin towards the recess of the retainer.

8. The system of claim 7, wherein the moment applied to the rocker arm by the spring pivots the rocker arm towards the retainer when weight of the float arm is removed from the rocker arm.

9. The system of claim 1, further comprising a retainer assembly, wherein the retainer assembly comprises: the retainer; a shaft pivotably coupled to the retainer; and a spring disposed on the shaft and configured to apply a moment to the retainer.

10. The system of claim 9, wherein the spring is compressed in response to engagement of the protrusion of the retainer by the cam, and wherein the compressed spring pivots the retainer about a second axis when the cam disengages the protrusion of the retainer.

11. A method for moving a cutterbar of an agricultural implement between a rigid configuration and a flexible configuration in response to articulation of a gauge wheel of the agricultural implement between a retracted configuration and an extended configuration, the method comprising: extending or retracting the gauge wheel of the agricultural implement; and simultaneously moving the cutterbar relative to a frame of the agricultural implement into the flexible configuration in response to retraction of the gauge wheel or simultaneously moving the cutterbar relative to the frame of the agricultural implement into the rigid configuration by an actuation system that is operated in response to extension of the gauge wheel; wherein the actuation system comprises: a rocker arm comprising a pin and pivotable about a first axis, the rocker arm coupled to a float arm; a retainer comprising: a recess that releasably receives the pin to selectively lock the pin into the recess; and a protrusion; and a cam coupled to the gauge wheel, the cam moveable with the gauge wheel assembly to selectively engage the protrusion of the retainer to move the retainer between a locked configuration and an unlocked configuration.

12. The method of claim 11, wherein simultaneously moving the cutterbar relative to the frame of the agricultural implement into the flexible configuration in response to retraction of the gauge wheel or simultaneously moving the cutterbar relative to the frame of the agricultural implement into the rigid configuration by an actuation system that is operated in response to extension of the gauge wheel comprises: extending the gauge wheel from the retracted position to the extended position; pivoting the cam in response to extension of the gauge wheel to disengage the cam from the protrusion of the retainer; and receiving the pin into the recess in response to disengagement of the cam from the protrusion of the retainer, thereby locking the cutterbar into the rigid configuration.

13. The method of claim 12, further comprising pivoting the rocker arm in a direction that moves the pin towards the recess of the retainer.

14. The method of claim 13, wherein pivoting the rocker arm comprises applying a moment to the rocker arm with a spring.

15. The method of claim 13, wherein pivoting the rocker arm in a direction that moves the pin towards the recess of the retainer comprises removing weight of the float arm from the rocker arm so that a moment applied to the rocker arm is great enough to pivot the rocker arm in the direction that moves the pin towards the recess of the retainer.

16. The method of claim 11, wherein simultaneously moving the cutterbar relative to the frame of the agricultural implement into the flexible configuration in response to retraction of the gauge wheel or simultaneously moving the cutterbar relative to the frame of the agricultural implement into the rigid configuration by an actuation system that is operated in response to extension of the gauge wheel comprises: retracting the gauge wheel from the extended position to the retracted position; pivoting the cam in response to retraction of the gauge wheel to engage the cam with the protrusion of the retainer; and releasing the pin from the recess in response to engagement of the cam with the protrusion of the retainer, thereby unlocking the cutterbar and placing the cutterbar into the flexible configuration.

17. An agricultural implement that automatically configures a cutterbar between a rigid configuration and a flexible configuration, the agricultural implement comprising: a frame; a gauge wheel assembly coupled to the frame and moveable between an extended position and a retracted position; a float arm pivotably coupled to the frame; a cutterbar engaged with the float arm, the cutterbar moveable between the rigid configuration and the flexible configuration in response to rotation of the float arm; and an actuation system that connects the float arm and the gauge wheel assembly, the float arm movable, via the actuation system, in response to retraction or extension of the gauge wheel to move the cutterbar between the rigid configuration and the flexible configuration, the actuation system comprising: a rocker arm comprising a pin and pivotable about a first axis, the rocker arm coupled to the float arm; a retainer comprising: a recess that releasably receives the pin to selectively lock the pin into the recess; and a protrusion; and a cam coupled to the gauge wheel assembly, the cam moveable in response to movement of the gauge wheel assembly to selectively engage the protrusion of the retainer to move the retainer between a locked configuration and an unlocked configuration.

18. The agricultural implement of claim 17, wherein extension of the gauge wheel assembly from the retracted position to the extended position rotates the cam to release the retainer to cause the retainer to pivot and capture the pin into the recess, placing the retainer into the locked configuration and the cutterbar in the rigid configuration.

19. The agricultural implement of claim 17, wherein retraction of the gauge wheel assembly from the extended position to the retracted position rotates the cam to engage the protrusion of the retainer, causing the retainer to release the pin from the recess, placing the retainer in an unlocked configuration and the cutterbar in the flexible configuration.
